Sagnik Bhattacharjee

Email | LinkedIn | GitHub

Education

Indian Institute of Engineering Science and Technology, Shibpur — Kolkata, India
BTech in Information Technology — *Expected July 2029*

Experience

Software Engineering Intern — Jan 2026 – Present
Dataring — *Remote*

- Implemented fusion-based aggregation in the risk engine and benchmarked it against MAX on synthetic workloads with weak-signal edge cases, reducing offline false negatives from **9.96% to 2.82%** (recall **90.0% to 97.2%**), recovering **71.7%** of weak violations; early production metrics are directionally consistent (limited volume).

Projects

Spheni – Vector Search Engine | *C++, Python, OpenMP* | GitHub — January 2026

- Architected a vector search library from scratch with **Python bindings (pybind11)**, implementing Flat and IVF-Flat indexes with **k-means++ clustering** and **inverted list storage**, achieving **12.6× speedup at 97% Recall@10** on 50k SIFT vectors through Voronoi-partitioned candidate scans.
- Diagnosed and eliminated **thread contention** in batch query execution by implementing **thread-local Top-K heaps with OpenMP dynamic scheduling**, scaling throughput from **2.4k to 5.7k QPS (2.37×)** while reducing p99 latency from **0.554 ms to 0.287 ms (1.93×)**.
- Reduced index memory footprint by **73%** (24.8 MB to 6.7 MB) through **INT8 scalar quantization**, trading **0.2% recall degradation** for efficient deployment on memory-constrained hardware.
- Designed benchmark harness measuring latency, throughput, and recall across nlist/nprobe sweeps; achieved **tight p99 tail latency** (within 1.5× of mean) through heap-based Top-K selection and unified cosine/L2 distance kernels.

Rayfloat – CPU Ray Tracer | *C++, OpenMP* | GitHub | Writeup — December 2025

- Architected a physically-based CPU renderer from scratch, implementing Lambertian, Metal, Dielectric, and DiffuseLight materials with an **iterative light transport model**, eliminating **recursion-induced stack pressure** and improving **Branch Target Buffer (BTB) predictability**.
- Diagnosed and eliminated a **41% performance penalty** from **cache-coherence contention** in `std::mt19937` by implementing a **thread-local XorShift32 pRNG**, restoring **near-linear multi-core scaling**.
- Reduced ray-scene intersection complexity from **O(n) to O(log n)** by designing a **Bounding Volume Hierarchy (BVH)** with **median-split construction** and optimized slab AABB tests, enabling scenes with 64,000+ primitives.
- Optimized **dielectric rendering** using **Schlick's Approximation** to numerically estimate the **Fresnel coefficient**, determining the probability of reflection vs refraction based on incident angles.

Amine – Productivity App | *Python, PyAutoGUI, SQLite* | GitHub | ProductHunt — August 2024

- Reached **#10 on ProductHunt (450+ downloads)** by building a multi-threaded distraction-blocking desktop app.
- Implemented system-level mouse boundary enforcement, keyboard interception, and full-screen tracking to prevent application switching during focus sessions.
- Designed an analytics dashboard using Matplotlib/Seaborn/Plotly with session insights and automated CSV export.
- Engineered **SQLite-backed session management** and input control, gaining **170+ GitHub stars**.

Skills

Languages: C++, Python, SQL
Systems: OpenMP, Linux, Git
Performance: Intel VTune Profiler, Perf, GDB
Web/Automation: FastAPI, Playwright